

WEEK 2

NumPy & Pandas for Data Analytics

PROJECT REPORT

Student	Tiwari Aman Dhananjay
Internship	Data Analytics Internship
Assignment	Week 2 — NumPy & Pandas
Duration	1 Week (7 Days)
Total Marks	100
Environment	Python • PyCharm • NumPy • Pandas

Practical Data Analytics Assignment

1. Assignment Overview

This report documents the practical work completed for the Week 2 assignment on NumPy and Pandas. The work progresses from basic numerical arrays to real dataset inspection, cleaning, transformation, aggregation and a complete mini data-analysis workflow.

10	Tasks Completed
100	Total Marks
4	Datasets Included
2	Core Libraries

Learning Objectives

- Work confidently with NumPy arrays, dimensions, indexing, slicing and reshaping.
- Perform mathematical and statistical calculations on numerical data.
- Create and manipulate Pandas Series and DataFrames.
- Read, inspect, filter, sort and clean CSV datasets.
- Handle missing values using removal and suitable imputation.
- Combine datasets and summarize information with GroupBy and Pivot Tables.
- Export processed data and build a small end-to-end analysis project.

Tools & Technologies

Python	NumPy	Pandas	PyCharm
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2. Task Roadmap

Task	Practical Work	Marks
01	NumPy arrays, shape, size, dtype and 1D/2D arrays	10
02	Indexing, slicing, row/column access and reshaping	10
03	Arithmetic + mean, median, min, max, std. deviation and sum	10
04	Pandas Series and Student DataFrame with new column	10
05	CSV loading, head, tail, shape, columns, dtypes, info, describe	10
06	Column/row selection, conditions and ascending/descending sort	10
07	Missing-value detection, removal and mean-based filling	10
08	Merge, concatenate, GroupBy aggregates and Pivot Table	15
09	Export processed DataFrame and verify CSV	5
10	Mini Student Performance analysis + insights + export	10

Submission Structure

The project package contains separate Python files for all ten tasks, reusable CSV datasets, generated output files, a README, a submission checklist and a dedicated screenshots folder for actual PyCharm evidence.

Task 1 — NumPy Introduction & Arrays

A numerical NumPy array containing ten values was created and displayed. The program also reports the array shape, size and data type, followed by examples of one-dimensional and two-dimensional arrays.

Key concepts: Core concepts: array creation • dimensions • shape • size • dtype

Task 2 — Indexing, Slicing & Reshaping

The program accesses individual elements using indexes, extracts a range using slicing, works with rows and columns of a two-dimensional array, and reshapes the original data into another dimension.

Key concepts: Core concepts: indexing • slicing • row/column selection • reshape

Task 3 — Mathematical & Statistical Operations

A numerical dataset is used for arithmetic operations and descriptive statistics. Addition, subtraction, multiplication and division are demonstrated together with mean, median, minimum, maximum, standard deviation and sum.

Key concepts: Core concepts: vectorized operations • descriptive statistics

Task 4 — Pandas Series & DataFrame

A Pandas Series is created first, followed by a student DataFrame. Column names and index information are inspected and a Result column is added to demonstrate DataFrame modification.

Key concepts: Core concepts: Series • DataFrame • columns • index • column creation

Task 5 — Reading & Inspecting Data

The Student Performance CSV is loaded with Pandas. The first and last five records, dimensions, column names, data types, dataset information and descriptive statistics are displayed.

Key concepts: Core concepts: read_csv • head • tail • shape • dtypes • info • describe

Task 6 — Selecting, Filtering & Sorting

The dataset is queried by selecting required columns and rows. Single-condition and multiple-condition filters are applied, followed by ascending and descending sorting based on Python marks.

Key concepts: Core concepts: selection • boolean filtering • multiple conditions • sorting

Task 7 — Handling Missing Values

A dedicated dataset containing missing values is used to identify null entries. The program demonstrates both removing incomplete rows and filling numerical missing values with the corresponding column mean.

Key concepts: Core concepts: isna • missing-value counts • dropna • imputation

Task 8 — Merge, Concatenate, GroupBy & Pivot Table

Two sales datasets are combined. A product-information table is merged using a common Product column, the sales tables are concatenated, category-level aggregates are calculated and a Region × Category Pivot Table is

produced.

Key concepts: Core concepts: merge • concat • GroupBy • aggregation • pivot_table

Task 9 — Exporting Data

The student dataset is enriched with Average_Marks and Performance fields, exported as a new CSV file and loaded again to verify that the processed data was saved correctly.

Key concepts: Core concepts: transformation • to_csv • verification

Task 10 — Mini Data Analysis Project

The Student Performance dataset is used for an end-to-end analysis. The workflow includes loading, inspection, missing-value handling, average-mark calculation, filtering high performers, sorting, department-level GroupBy analysis, a Department × Gender Pivot Table, key insights and final CSV export.

Key concepts: Core concepts: complete analytics workflow • insights • reporting

3. Dataset & Mini Project

The main analysis uses a Student Performance dataset with student identity, gender, department, age, subject marks, attendance and city. A second version intentionally contains missing numerical values for the missing-data task. Two sales datasets support the dataset-combination task.

Dataset	Purpose	Main fields
students.csv	Main analysis and inspection	Student_ID, Name, Gender, Department, Maths, Python, Database, Attendance, City
missing_students.csv	Missing-value handling	Same student fields with selected missing numerical values
sales_part1.csv	Sales operations	Order_ID, Product, Category, Region, Quantity, Unit_Price, Sales
sales_part2.csv	Sales operations	Order_ID, Product, Category, Region, Quantity, Unit_Price, Sales

Analysis Outputs

- Average_Marks is calculated from Maths, Python and Database scores.
- High-performing students are identified using an average-mark threshold of 90 or above.
- Department-wise average marks, average attendance and student count are summarized with GroupBy.
- A Pivot Table compares average marks across departments and genders.
- The program reports the highest-performing student, strongest department by average marks and department with the highest attendance average.
- The cleaned/processed student dataset is exported for reuse.

Why Missing-Data Handling Matters

Missing values can affect calculations, reduce the completeness of a dataset and produce misleading analytical results. The assignment therefore demonstrates both removal of incomplete records and filling numerical gaps with a simple column-mean method.

4. Evidence & Screenshot Checklist

The final submission requires actual program-output screenshots. The project package includes a screenshots folder so the evidence can be organized consistently.

Screenshot	What it should show
Task 1	Array + shape + size + dtype + 1D/2D arrays
Task 2	Indexing/slicing + original and reshaped arrays
Task 3	Arithmetic and statistical results
Task 4	Series + DataFrame + columns/index + updated DataFrame
Task 5	head, tail, shape, dtypes, info and describe
Task 6	Selections, filters and both sort directions
Task 7	Missing counts + before/after cleaning
Task 8	Merge, concatenate, GroupBy and Pivot output
Task 9	Processed DataFrame + export verification
Task 10	Analysis output + GroupBy + Pivot + key insights + export

Final Submission Checklist

- Run all ten Python files in PyCharm.
- Capture clear output screenshots.
- Place screenshots in the screenshots folder.
- Review the PDF and insert the screenshots in the evidence section.
- Upload the complete project folder to Google Drive.
- Set the Google Drive permission so the evaluator can open the files.
- Submit the PDF Report together with the Google Drive link.

Project Status	Code + datasets + report prepared. Actual PyCharm screenshots must be added by the student before final submission.
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5. Conclusion

This Week 2 assignment builds a practical foundation for data analytics with NumPy and Pandas. The work moves from numerical array operations to structured dataset analysis, cleaning, transformation, aggregation and export. The mini project combines the techniques into a reusable workflow that can be demonstrated through the accompanying Python source files, datasets and output evidence.

Prepared as a structured academic project report